

MODULE 3

Digital Sensor and Output Configuration and Industrial Simulation

PLC Siemens TIA Portal Course

Module 3: Digital Sensor and Output Configuration and Industrial Simulation

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Level: Basic

Software: Siemens TIA Portal

Language: Ladder Diagram (LAD)

1. Introduction

In industrial automation, sensors and digital outputs allow a PLC to interact with real machines and processes. Sensors feed information to the controller, while outputs activate devices such as motors, lamps, valves, and contactors.

Industrial simulation makes it possible to test programs before connecting them to a real machine, reducing errors and risks during commissioning.

2. Industrial Sensors

Sensors are devices capable of detecting physical changes and converting them into electrical signals the PLC can interpret.

Main types of sensors

Inductive sensor

Detects metal objects without physical contact.



Applications:

- Metal part counting
- Cylinder position
- Industrial limit stops

Characteristics:

- High durability
- Resistant to dust and oil
- Detects metal only

Capacitive sensor

Detects both metallic and non-metallic materials.



Applications:

- Liquid detection
- Plastic detection
- Level control

Characteristics:

- Detects multiple materials
- Sensitive to humidity and dirt

Photoelectric sensor



Detects objects using light.

Types:

- Through-beam
- Retro-reflective
- Diffuse

Applications:

- Conveyor belts
- Box counting
- Sorting systems

Limit switch



Mechanical switch that is physically actuated.

Applications:

- Motion limits
- Positioning
- Safety

3. Digital Inputs

Digital inputs receive binary signals:

- 0 = off
- 1 = on

Examples:

- Push buttons
- Sensors
- Switches

On Siemens PLCs they normally appear as:

I0.0

I0.1

I0.2

Where:

- I = Input
- The number represents the physical address

4. Digital Outputs

Digital outputs activate external devices.

Examples:

- Motors
- Pilot lights
- Relays
- Solenoid valves

Typical addresses:

Q0.0

Q0.1

Q0.2

Where:

- Q = Output

5. Configuring Sensors in TIA Portal

Basic configuration in TIA Portal consists of:

Step 1: Create the project

- Open TIA Portal
- Create a new project
- Add a PLC

Step 2: Configure inputs

Assign sensors to digital inputs:

Example:

Inductive sensor -> I0.0

Start button -> I0.1

Stop button -> I0.2

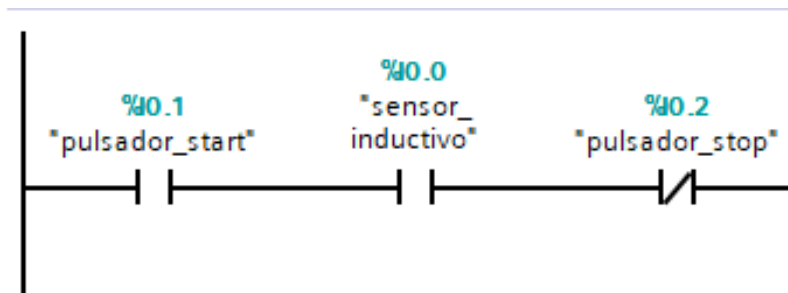


Diagram tags (original graphic, in Spanish): "sensor_inductivo" = inductive sensor · "pulsador_start" = start button · "pulsador_stop" = stop button.

Step 3: Configure outputs

Assign actuators:

Motor -> Q0.0

Green light -> Q0.1

Alarm -> Q0.2

6. Basic Ladder Example

System:

- Start button activates the motor
- Stop button stops the motor

Ladder program:

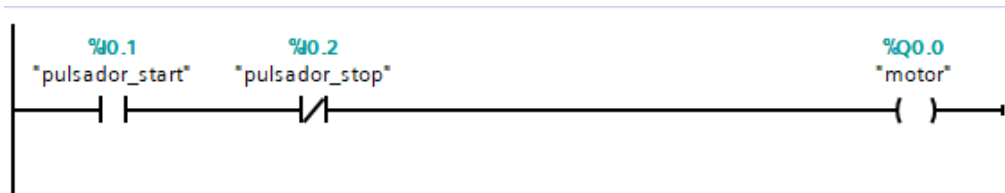


Diagram tags (original graphic, in Spanish): "pulsador_start" = start button · "pulsador_stop" = stop button · "motor" = motor.

Operation:

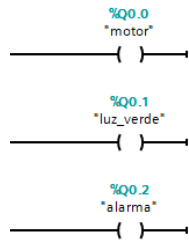
- I0.1 = Start
- I0.2 = Stop, normally closed
- Q0.0 = Motor

7. Industrial Simulation

Simulation allows testing the program without a physical PLC.

Siemens uses:

- PLCSIM
- PLCSIM Advanced



Advantages:

- Detect errors
- Test logic
- Save time
- Avoid real damage

8. Basic Use of PLCSIM

General steps

1. Compile the project

Check that there are no errors.

2. Start the simulation

Open PLCSIM from TIA Portal.

3. Download the program

Send the program to the virtual PLC.

4. Activate RUN

Switch the virtual PLC to RUN.

5. Test inputs

Activate virtual sensors:

I0.0 = TRUE

6. Check outputs

Observe activation:

Q0.0 = TRUE

9. Best Practices

- Name inputs and outputs correctly
- Use comments in Ladder
- Verify addresses before energizing
- Simulate before connecting real equipment
- Keep wiring and programming organized

10. Industrial Applications

Sensors and digital outputs are used in:

- Palletizing systems
- Conveyor belts
- Automatic machines
- Filling processes
- Production lines
- SCADA systems
- Industrial robotics

11. Conclusion

Correctly configuring sensors and digital outputs is fundamental in any automated system. Using industrial simulation makes it possible to validate PLC programs safely and efficiently before implementing them in real processes.

END OF MODULE 3

AUTOMATION STUDIO X

PLC Siemens TIA Portal Training System